Information Derived from PGSE-NMR ~Ion Diffusion Behavior, Molecular Association, Molecular Weight / Composition Correlation of Synthetic Polymers~

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Introduction

Pulse-Field Gradient Spin Echo NMR (PGSE-NMR) can be used to calculate self-diffusion coefficients of molecules. Since it is possible to determine the self-diffusion coefficient from each individual peak that can be observed in one-dimensional NMR spectra, we can evaluate the self-diffusion coefficients even for a mixture of multiple molecular species. We can also separate each signal by their diffusion coefficients using the DOSY method [1-4]. Further, it is an important point that the self-diffusion coefficients can be converted into information about the size and molecular weight of the molecules. This method also allows us to link the chemical composition information obtained from the one-dimensional NMR to the self-diffusion coefficients (molecular weight and molecule size) provided by PGSE NMR. Thus, these methods are expected to be very useful in a wide range of material analyses, as will be described below.

(1) Evaluation of ion diffusion in an electrolyte [5-8]
A principal function of an electrolyte is the conveyance of the ions. This conduction phenomenon can be evaluated using a direct physical parameter; the self-diffusion coefficient of the ion species. Even for a mixture of multiple ion species, the diffusion coefficients can be determined for each individual ion specie. For example, in the case of rechargeable lithium ion batteries, the movement of the lithium ions is particularly important, but this information cannot be obtained by assessing the ion conductivity measured for the dissociated cations and anions as a whole. Thus the information had been obtained using PGSE-NMR methods.

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(2) Evaluation of molecular association [9]
Upon association of molecules, the apparent molecular sizes will change, which should be observed as changes in the self-diffusion coefficients. Thus by applying DOSY method, molecular association can be detected.

(3) Evaluation of molecular weight / composition correlation [10-12]
The distribution of self-diffusion coefficients can be converted into the distribution of molecular masses. Furthermore, composition information obtained from the one-dimensional NMR spectra can be correlated with the diffusion coefficients, i.e. the molecular weight of the molecules. Thus, it is possible, for example, to use PGSE-NMR in applications where GPC-NMR is conventionally used, such as evaluating the correlation between the composition of a copolymer and its molecular weight

(4) Reaction tracking [13]
In situations where the size of the molecules changes, such as in synthesis reactions, reaction tracking can be performed by measuring the self-diffusion coefficients of the molecules using PGSE-NMR techniques. In contrast to GPC analysis, where the reaction must be stopped for analysis, this ability of PGSE-NMR to link the information of the self-diffusion coefficients with the information of chemical structures, obtained from the one-dimensional NMR spectra, eliminates the need to halt reactions midway to perform GPC analyses.

(5) Spectrum separation of mixtures
Even if the sample is a mixture of multiple components, separation of the spectra by diffusion coefficients using DOSY is possible. Thus, there is no longer any need to isolate the components in advance using HPLC or any other separation method. Combining DOSY with other two-dimensional NMR methods like COSY, enables a detailed structure analysis even for complex mixtures.

This is not a complete list, and there are innumerable applications of combined analysis of self-diffusion coefficients and chemical structure information. The ability of PGSE-NMR to relate the structure (composition), physical properties, and performance of a wide variety of materials holds great potential. In 2006, we had acquired a JEOL JNM-ECA400 series NMR spectrometer and a probe capable of generating gradient fields up to 13 T/m, and have been evaluating a wide variety of samples, from liquids to solids. Here we report several examples of our recent applications of PGSE-NMR for material analysis, including ion diffusion in electrolytes, molecular association, and assessment of the molecular weight / composition correlation of synthetic polymers.

Evaluation of Ion Diffusion in Electrolytes

The ion conductivity of an electrolyte is determined by the number and mobility of the dissociated ions. As indicated by the Nernst-Einstein equation, ion conductivity is a function of the self-diffusion coefficients and the sum of the number of cations and anions present, as well as the degree of dissociation.

Using PGSE-NMR, the self-diffusion coefficients can be determined separately for the anions and cations. This makes it possible to evaluate the individual contribution of the cations and of the anions to the ion conductivi-

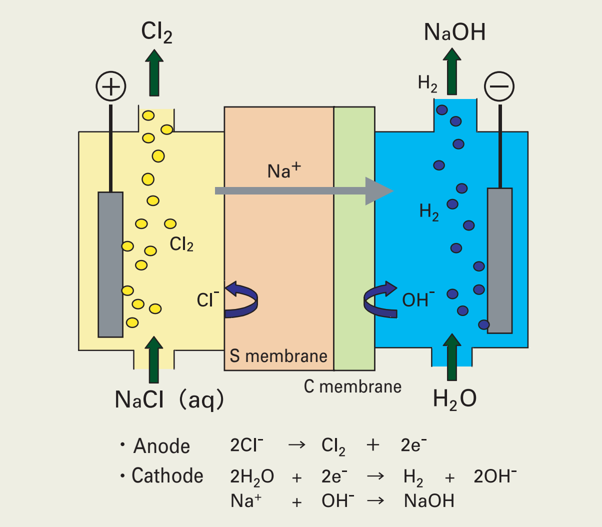

Fig. 1 Chlor-alkali electrolysis process.

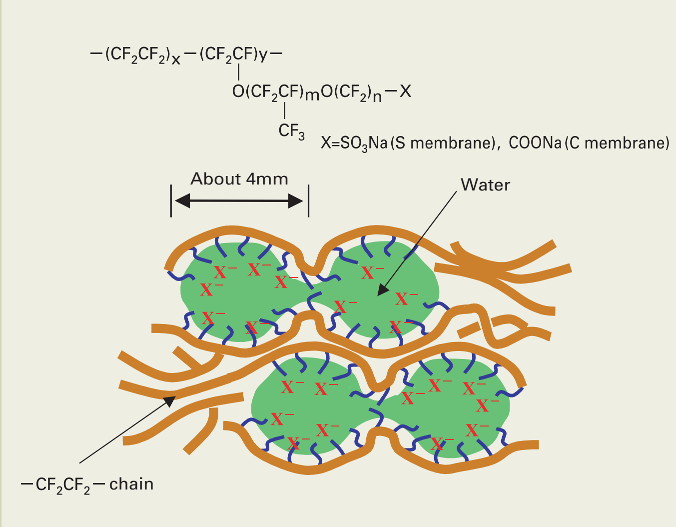

Fig. 2 Cluster structure and chemical structure of the perfluoro ion exchange membrane.

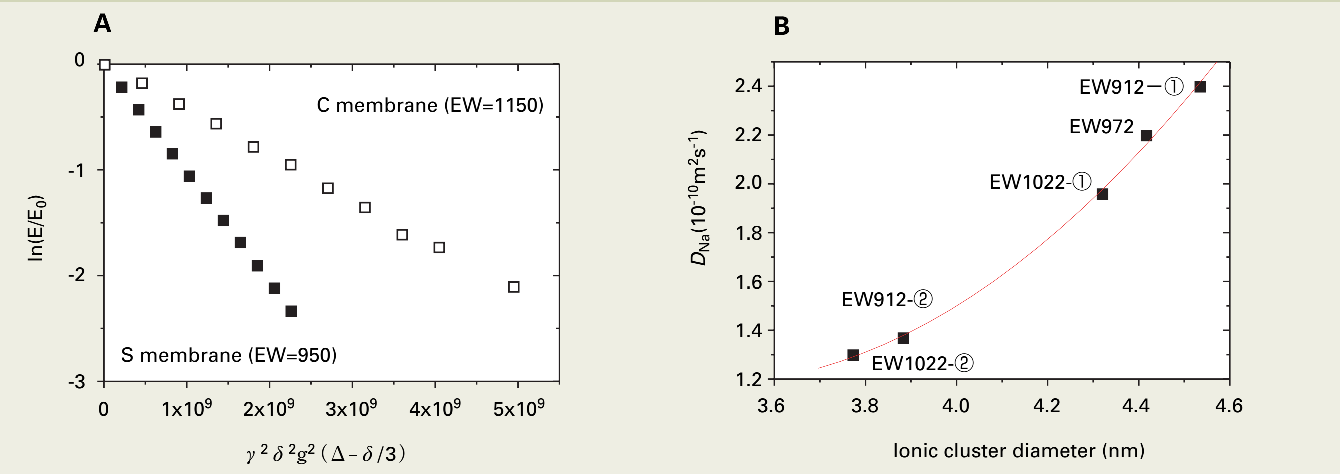

Fig. 3 (A) ^{23}Na - PGSE-NMR diffusion plot for the S membrane (EW = 950) and the C membrane (EW = 1150). S membrane: $D_{Na} = 1.0 \times 10^{-9}$ m^2/s, C membrane: $D_{Na} = 4.7 \times 10^{-10}$ m^2/s. Measurements were carried out with the membrane immersed in water at 90 °C.
(B) Relationship between the cluster diameter, determined using SAXS, and the Na^+ self-diffusion coefficient, determined using PGSE-NMR, for membranes with different EW. Both SAXS and PGSE-NMR were performed with the membrane immersed in water at room temperature. ① and ② are polymers with the same EW, but having different membrane formation conditions to control the cluster diameter. In both cases, a PGSE Spin Echo (PGSE-SE) sequence was used for the measurements.

ty, which cannot be obtained by measuring the ion conductivity alone. In this way, it is possible to extract information about the degree of dissociation.
Here, we present diffusion data of Na^+, Cl^- in chlor-alkali electrolysis, water in fuel cells, and Li^+ in a lithium ion battery.

Chlor-Alkali Electrolysis [14]

Chlor-alkali electrolysis is the process of producing chlorine and sodium hydroxide from NaCl aqueous solution by a membrane electrolysis (**Fig. 1**). As shown in **Fig. 2**, a fluorine-based electrolyte membrane, with a carboxylic acid group (C membrane) and a sulfonic acid group (S membrane), is commonly used. Due to the amphipathic nature, the exchange groups within the membranes form water-containing micelles (clusters), inside which the ions move. An important characteristic required for the membranes is the selective permeability of Na^+ cations and the blocking of anions. The structure of these clusters may have an influence on the cation selectivity. We comprehensively evaluated the cluster structure using small-angle x-ray scattering (SAXS), and the self-diffusion coefficients of the cations and anions using PGSE-NMR.

Since both T_1 and T_2 are short (10 ms or less) for ^{23}Na, a long diffusion time (Δ) could not be used in the diffusion measurement. Furthermore, due to the small gyromagnetic ratio, it was necessary to apply a large magnetic field gradient to attenuate the signals. By using a 13 T/m gradient magnetic field probe, we were able to obtain a good ^{23}Na PGSE-NMR diffusion plot (**Fig 3**A).

Distinct difference were observed between the S membrane and the C membrane with regard to Na^+ diffusion (S membrane: $D_{Na} = 1.0 \times 10^{-9}$ m^2/s, C membrane: $D_{Na} = 4.7 \times 10^{-10}$ m^2/s). This small Na^+ self-diffusion coefficient in the C membrane was a result of the low water content and the small cluster diameter in the membrane. Using membranes for which the cluster diameter was controlled by changing the film-forming conditions and equivalent weight (EW), the relationship between the cluster diameter (determined using SAXS) and the Na^+ diffusion coefficient (determined using PGSE-NMR) was clarified (Fig. 3B). It is clear that as the diameter of the clusters that form the path for Na^+ becomes larger, the speed of the Na^+ diffusion increases. The non-linearity in this relationship might be due to the fact that the increase in the cluster diameter does not only make the ion path larger, but also has an effect on the formation of networks and coupling among individual clusters. Furthermore, direct proportional relationship between these Na^+ diffusion coefficients and the ion conductivity were observed. This means that the Na^+ ion conductivity is in accordance with the previously mentioned Nernst-Einstein equation. This is in contrast to the fact that the observed relationship between the ion conductivity and the diffusion coefficient of water for the fuel cell membranes is not linear. This will be discussed later.

In order to evaluate the cation selectivity, we evaluated the diffusion of Na^+ (^{23}Na-NMR),

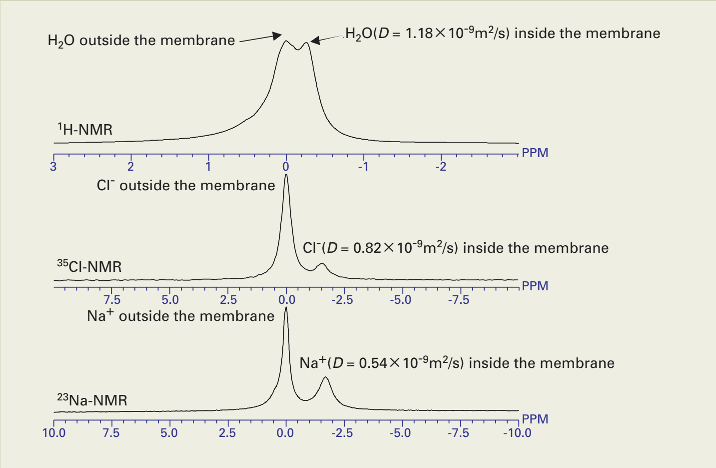

Fig. 4 Self-diffusion coefficient evaluated using PGSE-^{23}Na-, ^{35}Cl-, ^{1}H-NMR. Measurements were performed with membranes immersed in 3.5N NaCl aqueous solution at 80 °C.

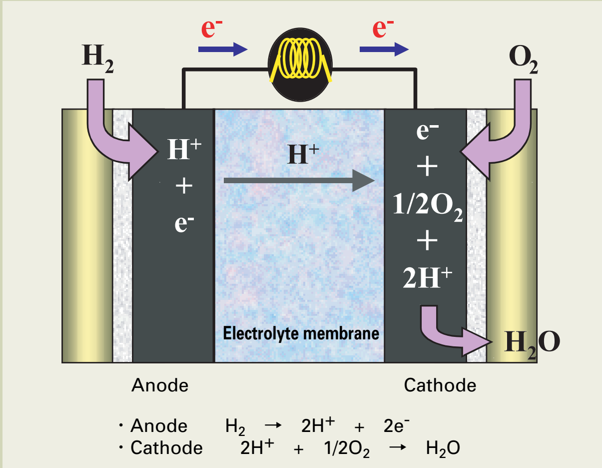

Fig. 5 Solid polymer fuel cell.

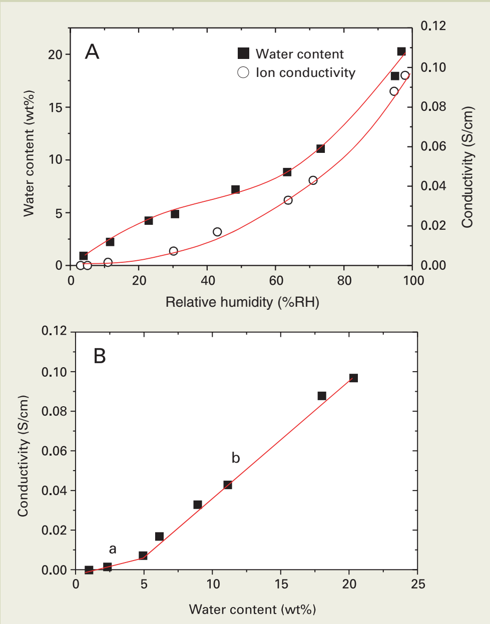

Fig. 6 (A) Dependency of the water content and ion conductivity of the electrolyte membrane on relative humidity.
(B) Relationship between water content and ion conductivity. Membrane: S membrane (proton type), EW = 950.

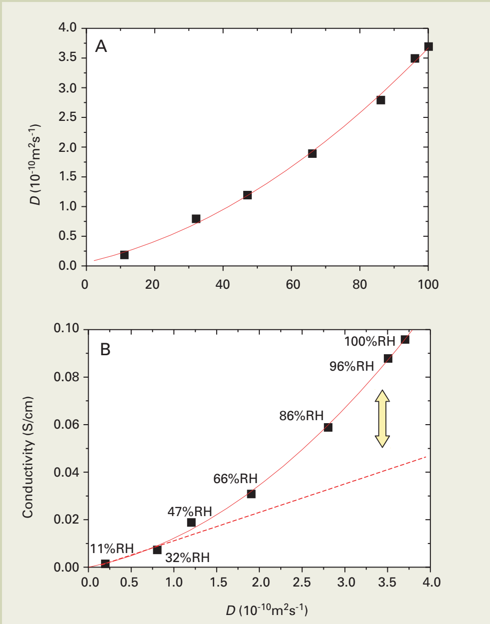

Fig. 7 (A) Dependency of the self-diffusion coefficient of water on relative humidity.
(B) Relationship between the self-diffusion coefficient of water and ion conductivity. Membrane: S membrane (proton type), EW = 950, measured at room temperature.

Cl^- (^{35}Cl-NMR), and H_2O (^{1}H-NMR) using an S membrane soaked in NaCl (80 °C). ^{35}Cl is a low-frequency nucleus, and is outside the frequency range of a conventional probe (JEOL Ltd. TH5 probe), but we managed to perform the measurements by adjusting the tuning beyond the normal limit. The chemical shifts of the ions inside and outside the membrane were different in the spectra, and we were able to selectively evaluate the diffusion of the ions in the membrane (**Fig. 4**).

These results can be directly related to the electrolyte performance; such as the membrane resistance (Na^+ diffusion coefficient), cation selectivity (Na/Cl diffusion coefficient ratio), and the amount of water permeation (H_2O diffusion coefficient). The ion conductivity measurement can not give each parameter individually. This demonstrates the power of the PGSE-NMR method.

Polymer Electrolyte Fuel Cells [15]

An outline of a Polymer Electrolyte Fuel Cell (PEFC) is shown in **Fig. 5**. The polymers commonly used as the exchange membrane are sulfonate membranes. In PEFC, the H^+ in the membrane move from the anode to the cathode, accompanying water molecules. It is also known that the ion conductivity of the original fluorine type membranes depends on the relative humidity (**Fig. 6**A), with a rapid drop of the ion conductivity observed under low humidity. Therefore, it has been necessary to operate PEFC with the membranes in a high-humidity environment.

The relationship between ion conductivity and water content is shown in Fig 6B. As the water content becomes higher, the ion conductivity increases, revealing the importance of the presence of water. Furthermore, there is an inflection point in the graph of this relationship, indicating the presence of two types of water; type **a**, which makes a small contribution to the ion conductivity, and type **b** with a large contribution. Thus, it was crucial to analyze the states of the water to understand the proton conduction phenomena.

Then we measured the diffusion coefficient of water under various humidity conditions in order to clarify the dependency of proton conductivity on humidity, and the results were analyzed along with information of the states of the water and the membrane cluster structure obtained from IR and SAXS measurements. All the PGSE-NMR, SAXS and IR measurements were performed *in-situ* while controlling the humidity. To achieve this, a saturated brine solution was sealed into the sample measurement cells. Since the saturation vapor pressure varies according to the type of salt, it is possible

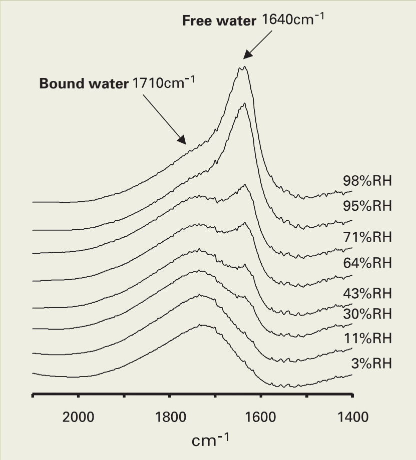

Fig. 8 IR spectra measured on varied relative humidity.
Membrane: S membrane (proton type), EW=950, measured at room temperature.

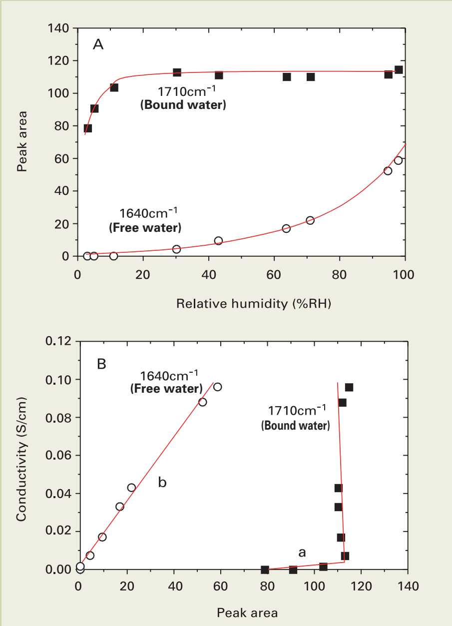

Fig. 9 (A) IR peak area for free and bound water at varied relative humidity.
(B) Relationship between IR peak area and the ion conductivity. Membrane: S membrane (proton type), EW = 950.

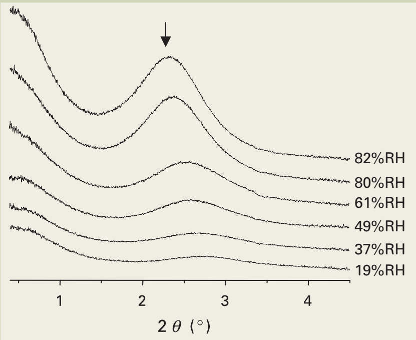

Fig. 10 SAXS profile measured at varied relative humidity.
Membrane: Sulfonate membrane (proton type), EW = 950, measured at room temperature.

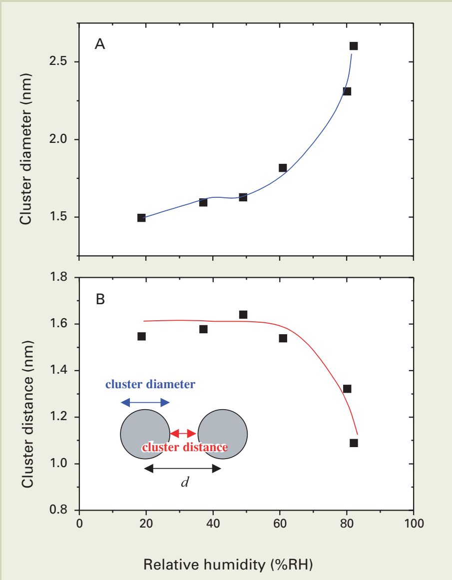

Fig. 11 The dependency of the cluster diameter determined using SAXS on relative humidity (A), and of the inter-cluster distance (B).
Membrane: S membrane (proton type), EW = 950.

to control the humidity inside the sample cell. The salts used include LiCl-H_2O (11% RH), $CaCl_2$-$6H_2O$ (32% RH), KNCS (47% RH), $NaNO_2$ (66% RH), NaCl (76% RH), KCl (86% RH), and K_2SO_4 (96% RH).

Fig. 7A shows the dependency of the self-diffusion coefficient of water in the membrane on humidity. As the humidity lowers, there is a rapid decrease in the diffusion coefficient of water, revealing that this is the source of the drop in the ion conductivity. Fig. 7B shows the correlation between the self-diffusion coefficient of water and the ion conductivity. In contrast to the previously mentioned direct linear relationship exhibited between the ion conductivity and the Na^+ diffusion coefficient for chlor-alkali electrolysis, the relationship in this case is represented by a curve. As the humidity rises, it appears that the increase in ion conductivity becomes greater than the rise of the diffusion coefficient.

To further investigate the source of this change, the humidity dependency of the results of IR and SAXS were investigated. In the IR spectrum, two types of peaks were observed; in positions 1640 cm^{-1} and 1710 cm^{-1} (**Fig. 8**). The 1710 cm^{-1} peak corresponds to the water directly bonded to the sulfonate proton by hydrogen bonding (bound water). The 1640 cm^{-1} peak is assigned to the free water demonstrating the same behavior as bulk water [16]. Under conditions of low humidity, the amount of bound water increases rapidly as the humidity rises (up to about 30% RH). As the humidity rises above this level, there is no increase in the bound water (**Fig. 9**). In comparison, the free water steadily increases above 30% RH, with a rapid increase appearing at levels over 70% RH. The relationship between the ion conductivity and the amounts of free water and bound water (IR peak areas) (Fig 9B), reveals that the increase in bound water at low humidity (see **a** in Fig 9B) only makes a slight contribution to the ion conductivity, while the growth of free water makes a large contribution to the increase in ion conductivity (see **b** in Fig 9B). This suggests that the water types **a** and **b** in Fig 6 discussed earlier originate from the bound and free water, respectively.

The cluster structure was analyzed using SAXS. The obtained peaks (indicated by the arrow ↓ in **Fig 10**) were assigned to the scattering interference peaks for each cluster. The cluster distance (*d*) was determined from the scattering angle using the Bragg's formula. Furthermore, the cluster diameter was determined by assuming the clusters to be rigid-body spheres and performing fitting. The dependency of the SAXS profile on the relative humidity is shown in Fig. 10. It is clear that for this electrolyte membrane the cluster structure exhibits an extremely dynamic alternation dependent on the external environment. The increase in the peak intensity at high humidity may be due to the increase of the water content. Also, a shift of the peak to the low-angle side was observed, suggesting the growth of *d*. **Fig 11**A shows the cluster diameter, determined from the fitting with the SAXS profile dependent on relative humidity. We can observe a rapid increase in the cluster diameter at relative humidity of over 70% RH. Further, the value obtained by subtracting the cluster diameter from *d* (= separation between adjacent clusters) drops drastically as the relative humidity increases, revealing that the individual clusters are in close proximity (Fig 11B). These rapid changes at around 70% relative humidity correspond to the region of rapid increase in free water discussed earlier.

These results observed for the states of the water support the following conclusions with regard to the non-linear relationship between the ion conductivity and the diffusion coefficient shown in Fig 9B, and the dependence of the H^+ conduction mode on the relative humidity.

- In the range of 0 -30% RH, the water is directly bonded to the sulfonic acid group's proton by hydrogen bonding. This bound water has a low mobility and does not make a large contribution to the ion conductivity.
- In the range of 30 -70% RH, there is no increase in the bound water, and only a slight increase in free water. The free water makes a large contribution to the ion conductivity. However, since the amount of increase is small, there is no large increase in ion conductivity.
- In the range of 70 -100% RH, there is a sudden jump upward in the ion conductivity accompanying a rapid increase in the free water. We can see that the ion conductivity increases more than the corresponding rise of the diffusion coefficient in this range.

At humidity levels below 60% RH the ion conduction phenomenon appears to be the H^+ movement accompanying water (Vehicle mechanism). In contrast, at humidity levels of over 70% RH, there is an increase in the free water and the diameter of the clusters makes the path for the ions larger. In addition, the individual clusters form a network, so it is believed that the ion conductivity is manifested as the H^+ hopping between water molecules (Grotthus mechanism). This may be the cause of the observed increase in ion conductivity exceeding the contribution of the diffusion coefficient under conditions of high humidity.

As demonstrated above, PGSE-NMR does not simply offer information on the self-diffusion coefficient; the results can be combined with those from IR and SAXS to perform a comprehensive analysis, even enabling interpretation of the proton conduction mechanism, making PGSE-NMR an extremely useful tool.

Rechargeable Lithium Ion Batteries [17]

The movement of the Li^+ ions is important for rechargeable batteries, but it is only possible to evaluate the total mobility of all ions (cations and anions) with ion conductivity measurements. PGSE-NMR can be used to evaluate the Li^+ and the anions separately, and calculations can be performed to determine the Li^+ transport number and degree of dissociation. Furthermore, as presented in the following example, it is possible to analyze not just the ion self-diffusion coefficients, but also the diffusion behavior. **Fig. 12** shows the diffusion plots for the self-diffusion coefficient of Li^+ within two types of polyketone electrolyte membranes with the varying diffusion time (Δ). The self-diffusion coefficient of Li^+ within the Type 1 electrolyte membrane does not exhibit any dependency on the diffusion time. In contrast, the Li^+ diffusion in the Type 2 membrane shows a clear dependency on the diffusion time. The value of the Li^+ self-diffusion coefficient was $D_{Li} = 5.6 \times 10^{-11}$ $\Delta m^2/s$ for Type 1, and $D_{Li} = 0.4 \times 10^{-11}$ m^2/s (Δ = 500 ms) for the Type 2 membrane; so, the value of the Type 1 electrolyte membrane was more than 10 times greater than that of Type 2. Type 1 was an amorphous membrane, while Type 2 was a crystalline membrane. And because of this difference the two should show different diffusion behavior in which the amorphous material (Type 1) should show free diffusion of Li^+ and Type 2 material should pass through barriers, such as crystal sites. This may explain the non-uniform diffusion indicated by the self-diffusion coefficient becoming smaller as the diffusion time becomes longer. In this way, the ability of PGSE-NMR to obtain information on the ion diffusion behavior as well as the self-diffusion coefficients makes it possible to establish additional design parameters for membranes.

Evaluation of Molecular Association

The association of molecules should alter the apparent molecular size. And the self-diffusion coefficient is inversely proportional to the Stokes radius of the molecule. Therefore, it should be possible to detect the association by monitoring the self-diffusion coefficients. Here, we present an example of an analysis of a cyclodextrin inclusion complex.

Evaluation of Cyclodextrin Association [18]

Cyclodextrin (CD) is a cyclic oligosaccharide composed of α 1-4 linked glucose molecules. A six-membered sugar ring is called αCD, a 7 sugar ring molecule, βCD, and the 8-sugar ring molecule, γCD. The outside region of the CD ring is hydrophilic, while the interior is hydrophobic, and this feature gives CD the property of trapping and retaining a variety of compounds (guest molecules) inside its ring [19, 20] (**Fig. 13**). These inclusion complexes are widely used in the pharmaceutical, food product and chemical saccharide fields, taking advantage of the solubility, extended-release and masking effect properties.

Conventionally, the detection of the association of a CD host and a guest molecule was preformed by methods like tracking changes in UV or 1-dimensional NMR spectra. The difficulty, however, is that these changes in the spectra do not necessarily indicate the inclusion directly. In comparison, a DOSY method allows us to obtain parameters that are related to the apparent molecular size, i.e. the diffusion coefficients, which should enable a more direct evaluation of the inclusion phenomenon. Upon association, the diffusion coefficient of the host and guest molecules should become identical.

First, we present an example of an analysis of the inclusion of para-nitrophenol (PNP) in αCD. DOSY measurements were performed against a sample mixture of these molecules in a 1:1 (mole ratio) in water (**Fig. 14**B). The results show that although the diffusion coefficient of PNP is smaller than that of PNP alone in an aqueous solution, it is not equal to that of αCD. Then, the mixture ratio of these two compounds was varied and DOSY measurements were performed. The results show that the diffusion coefficient of PNP was dependent on the mixture ratio; as the PNP fraction decreases, the value approaches the αCD diffusion coefficient (Fig. 14 A, B, C). This suggests that in the time scale of DOSY measurements, a rapid exchange occurs between the free and CD associated PNP. Thus the average self-diffusion coefficients of the free and included PNP was observed. As the proportions of PNP and αCD were varied, the proportion of PNP captured in the αCD also changed, causing the alternation of the diffusion coefficient. At the lower PNP content, the percentage that was captured by the αCD was larger, and the observed value approached the αCD diffusion coefficient. In this way, by analyzing the variation in the diffusion coefficient relative to the PNP/CD ratio it is also possible to obtain information about the association constant. We also note that the self-diffusion coefficient of αCD was independent of changes in the mixture ratio. This is due to the fact that there is no significant change in the Stokes radius of αCD, even when associated with PNP.

Next, an example of the inclusion of 4,4'-biphenyl dicarboxylic acid sodium salt (BDC) in αCD will be shown. As can be seen from **Fig. 15**A, a diffusion coefficient value observed for BDC agrees with that of αCD. In addition, there are also peaks obtained that match the diffusion coefficients for the BDC and the αCD alone. This should correspond to the free molecules that are not involved in an inclusion complex. This indicates that unlike PNP, BDC is stably contained in the inclusion complex (does not show fast exchange). ^{1}H-NMR peaks corresponding to the CD-BDC complex was assigned by their diffusion coefficient. Then the molar proportion of BDC and αCD was clearly identified to be 1:2. The diffusion coefficient for the inclusion complex is smaller than that of the free αCD, which also suggests that the complex is not simply a 1:1 pairing of BDC and αCD; rather, a dimer

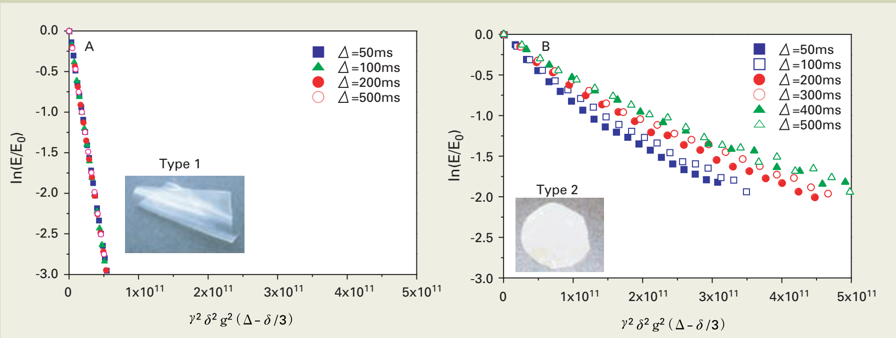

Fig. 12 Diffusion plot of ^{7}Li –PGSE-NMR measured at varied diffusion times for polyketone-type electrolyte membrane.
(A) Type 1 electrolyte membrane: $D_{Li} = 5.6 \times 10^{-11}$ m^2/s
(B) Type 2 electrolyte membrane: $D_{Li} = 0.4 \times 10^{-11}$ m^2/s (Δ = 500 ms)

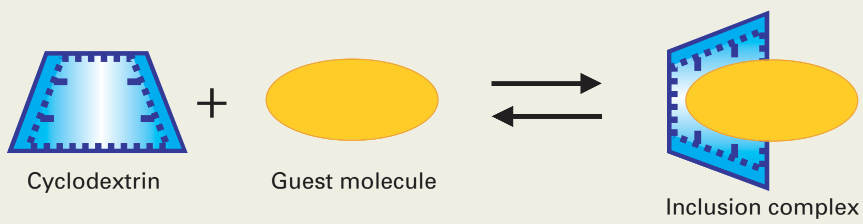

Fig. 13 Inclusion of a guest molecule in CD.

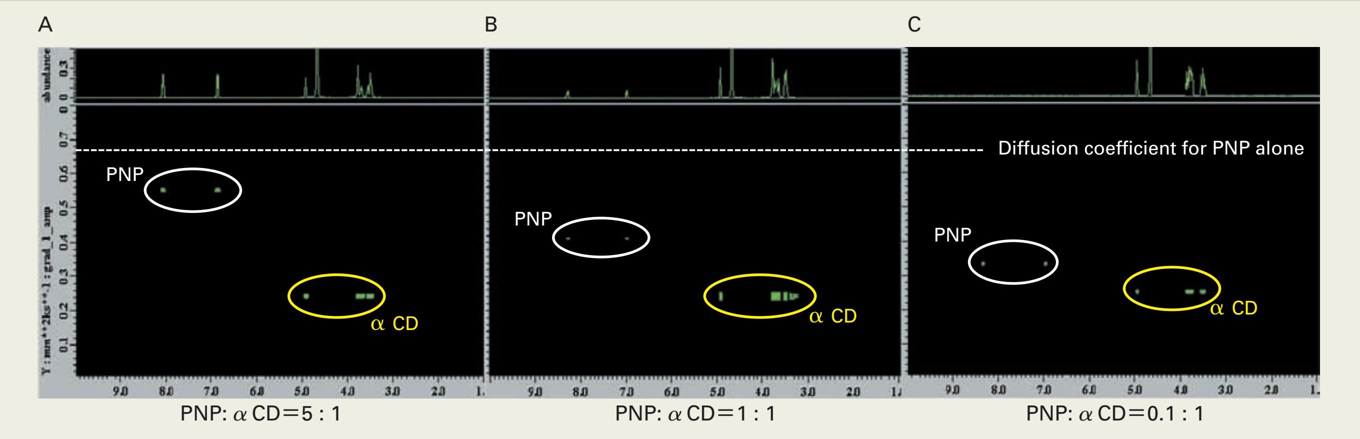

Fig. 14 ^{1}H-DOSY spectrum for a PNP-αCD inclusion complex. (A) PNP: αCD = 5:1 (mole ratio), (B) PNP: αCD = 1:1 (mole ratio), (C) PNP: αCD = 0.1:1 (mole ratio). DOSY measurements were performed at room temperature.

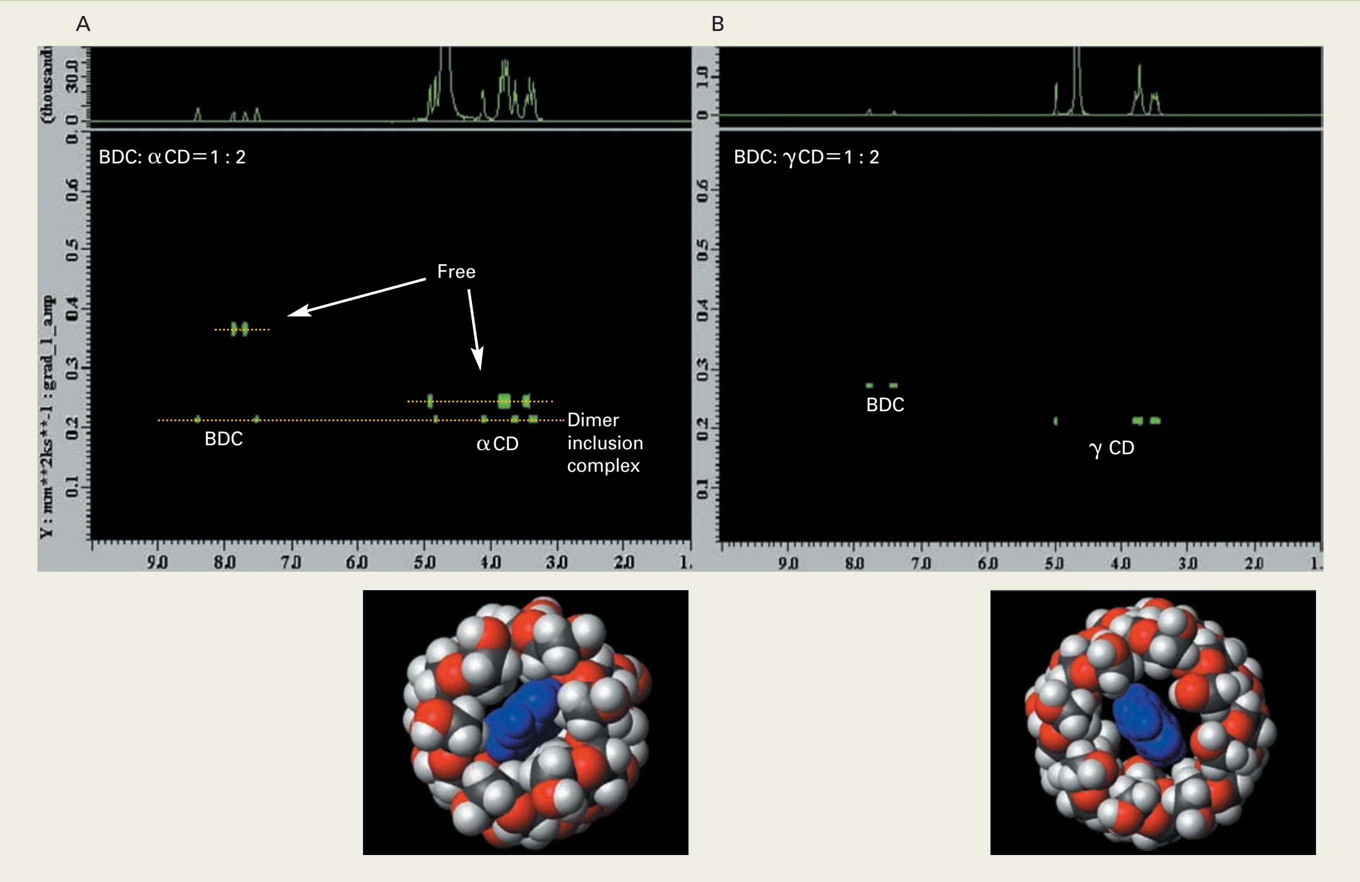

Fig. 15 ^{1}H-DOSY spectra. (A) BDC: αCD = 1:2 (mole ratio), (B) BDC: γCD = 1:2 (mole ratio).

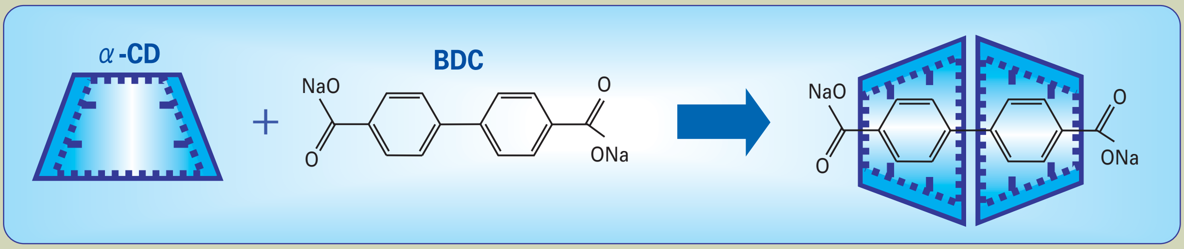

Fig. 16 Formation of the BDC and αCD dimer inclusion complex.

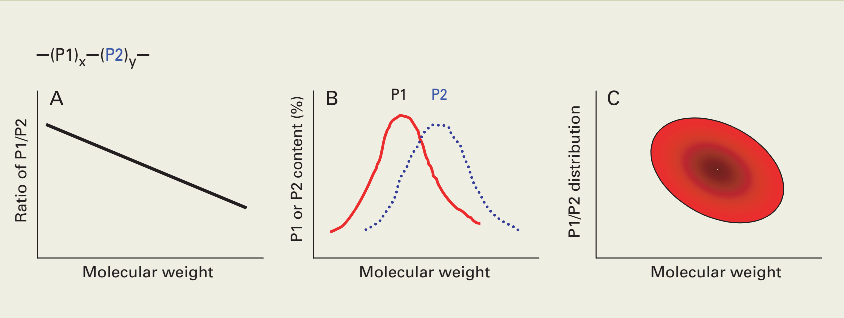

Fig. 17 Schematic drawing of the distribution of copolymer composition dependency on molecular weight.

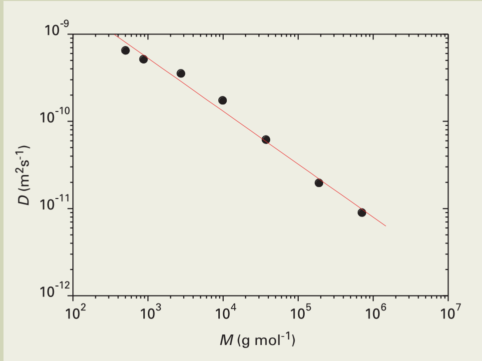

Fig. 18 Relationship between the molecular weight of polystyrene (M) and the self-diffusion coefficient (D). $D = (2.1 \times 10^{-8}) \times M^{-0.6}$

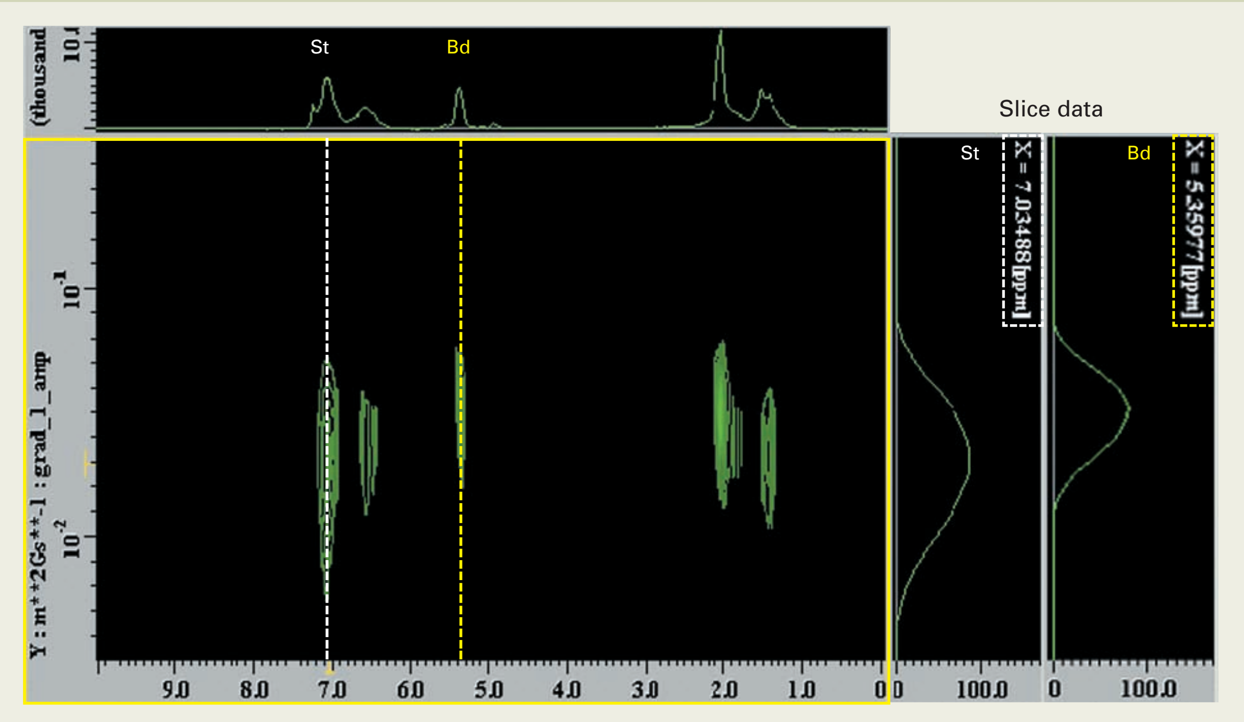

Fig. 19 ^{1}H-DOSY spectrum of styrene-butadiene copolymer. The sample was dissolved in $CDCl_3$, and measurements were performed at room temperature.

inclusion complex, as illustrated in **Fig. 16**. This dimer structure is thought to be the reason for the stability of this inclusion complex. The trapping of the BDC molecules by 2 molecules of αCD in the solution also agrees with the results from X-ray structure analysis of the crystal [21].

In contrast, in the case of the BDC-γCD system, the diffusion coefficients do not match between BDC and γCD (Fig. 15B), unlike the BDC-αCD system. In the same way as observed for PNP-αCD, there appears to be rapid exchange in the time scale of the DOSY measurements. Since the inner diameter of γCD is large for the BDC, so the molecules do not fit precisely, and a stable complex is not formed.

We have presented an example of an analysis of CD association using DOSY, and have shown that the intuitive and straightforward assessment of molecular association is capable. Furthermore, this method also offers the ability to acquire comprehensive information such as the association constants, stability (exchange), and the inclusion structure. In addition, because the time scale for the DOSY measurements is dependent on the diffusion time (Δ), as discussed for the evaluation of lithium diffusion above, by freely varying the time scale of the measurements, and tracking the Δ-dependence of the diffusion coefficient, it is possible to obtain information on the exchange rate of the complex.

Evaluation of Correlation between Molecular Weight and Composition of Synthetic Polymers

In the case of a synthetic copolymer composed of several monomer constituents, the composition ratio of the monomers is often used as a parameter to specify the structure of the polymer. However, the molecular mass dependency (**Fig. 17**A, B) or distribution (Fig. 17C) of this composition is also an important parameter for the polymer properties. Thus, there is a strong demand for detailed analysis of the composition and structure. For example, for the molecular mass dependence of the composition, the conventional technique had been used to perform NMR analysis on each of the GPC fractions, or to apply a GPC-NMR method in which GPC and NMR are linked on-line. However, not only are these methods complicated and cumbersome, these methods also have the disadvantages that once an fractionated portion has been dried, it can no longer be dissolved in the solvent; or, as for GPC-NMR, it is necessary to use a costly NMR solvent as the eluent. In comparison, with DOSY the self-diffusion coefficient and its distribution can be determined for each individual peak from each component. Therefore, evaluation of the correlation between the molecular weight (diffusion coefficient) and the composition (NMR spectrum) is possible. Here we present an example of analysis of a styrene-butadiene copolymer using DOSY.

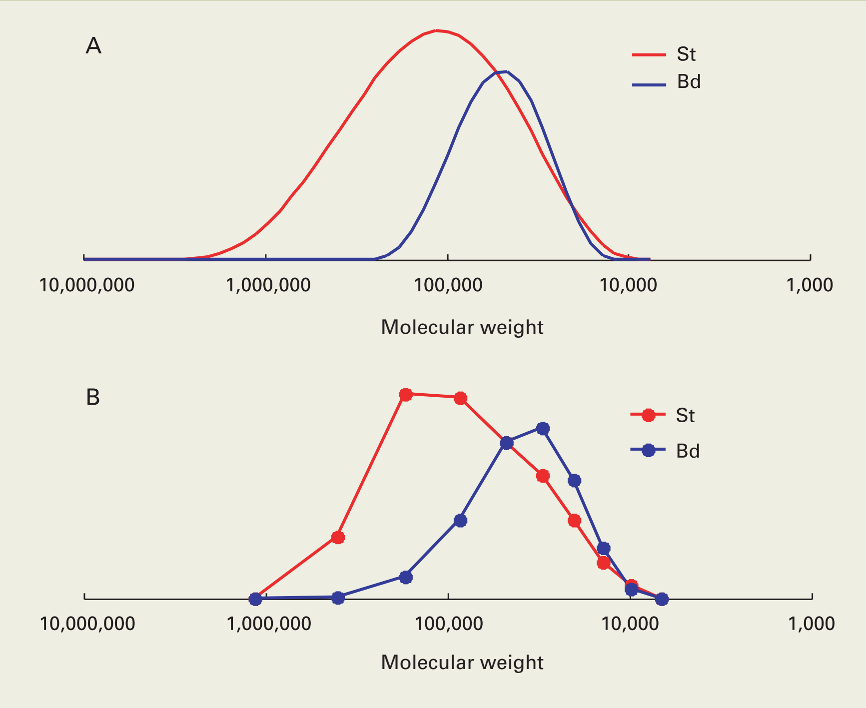

Fig. 20 Dependency on molecular weight of the styrene and the butadiene unit content.
(A) As determined from the slices of the DOSY spectra of the styrene peak (7.1 ppm) and the butadiene peak (5.4 ppm). CONTIN protocol was used for the inverse LaPlace transform.
(B) As determined from the styrene / butadiene ratio from the ^{1}H-NMR of the GPC fraction (conventional method)

Correlation between Composition and Molecular Weight for Styrene-Butadiene Copolymer

The relationship between the molecular weight and the diffusion coefficient for standard polystyrene of differing molecular weights (usually used for GPC molecular weight calibration) is shown in **Fig. 18**. On the log-log plot, a nearly straight line is obtained. Using this relationship as the standard curve, calculation of the molecular weights from the obtained diffusion coefficients is possible. However, in the same way as the molecular weight calibration curves for GPC, when it is not possible to generate the calibration curve using the identical polymer as in the test sample, it is difficult to determine the true molecular weight. Therefore, it is necessary to keep in mind that the results will be a "polystyrene-equivalent weight", in the same way as for GPC.

We used DOSY on a styrene-butadiene copolymer to determine the dependency of the styrene / butadiene unit composition on the molecular weight (**Fig 19**). The 7.1 ppm aromatic ring peak for the styrene unit, and the 5.4 ppm double-bond peak of the butadiene unit were used to obtain diffusion coefficients and the distribution. Using the calibration curve mentioned above, the molecular weight and its distribution were obtained for styrene and butadiene units individually. The results of the DOSY experiment were compared to the results of the conventional method in which NMR composition analysis were performed for each GPC fraction.

The comparison of the results from the conventional method (GPC fraction → NMR) with the DOSY results is shown in **Fig. 20**. The results from these two methods are in close agreement, confirming the reliability of the DOSY method. It is clear that the high molecular weight components are styrene unit rich, with the components having a molecular weight of about 1,000,000 being nearly 100% styrene. In contrast, the low molecular weight components in the range of 100,000 to 300,000 have styrene and butadiene units in nearly equal proportions.

Applying the DOSY method in this way enables us to make detailed characterizations of a synthetic polymer. It is not only significantly simpler than the conventional methods, it is especially effective for analyzing compounds that become denatured when the fractions are concentrated and dried. For the styrene-butadiene copolymer used for this study, once the GPC fraction solution is dried, it cannot be re-dissolved, so care is required in the concentration process. DOSY does not have this disadvantage.

DOSY provides information about the dependency of the composition on the molecular weight, as shown in Fig. 17 A and B. The analysis of the composition distribution, however in Fig. 17 C is not possible. For such a case, a two-dimensional GPC-LC method, composed of a combination of GPC (molecular weight isolation) and LC (composition isolation), is required. The drawbacks are that it is not easy to define the LC conditions, and it is necessary to search for the optimal conditions for each sample. Therefore, it is necessary to consider and select the best method according to the information that is required.

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